

Algebra First-Year Exam, May 2014, Part 2

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. Let $R = \mathbb{Z}[i]$ be the ring of Gaussian integers. Let M be a maximal ideal of R . Prove that R/M is a finite field of order p or p^2 where p is a (rational) prime number.
2. This problem has three parts.
 - (a) Give an example of an integral domain which is not a unique factorization domain.
 - (b) Give an example of a principal ideal domain which is not a field and is not isomorphic to $\mathbb{Z}$.
 - (c) Give an example of a commutative ring with identity which is not an integral domain.
3. This problem has two parts.
 - (a) Define Euclidean domain.
 - (b) Prove that every Euclidean domain is a principal ideal domain.
4. Let R be a ring with identity and let M be an R -module. Prove that the annihilator $\text{Ann}(M)$ is a two sided ideal of R .
5. Let V be a vector space of dimension 4 over the field $\mathbb{Q}$ of rational numbers. Let $T : V \rightarrow V$ be a linear transformation. Suppose that $(T^2 - \text{Id})^2 = 0$.
 - (a) What are the possible minimal polynomials for T ?
 - (b) For each possible minimal polynomial from (a), give a matrix with this minimal polynomial.
6. Let F/K be an extension of fields, and suppose that $[F : K] = 5$. Prove that there exists an irreducible polynomial $p(x) \in K[x]$ such that F is isomorphic to $K[x]/(p(x))$.